

ARM-450 — 2D Drawing Package

- Project: ARM-450, 6-DOF articulated arm, 450 mm overall
- Date: 2026-08-19
- Units: millimetres. Third-angle projection.
- Source: every view is projected from the exported CAD solid, not sketched.

1. Assembly specification

parameter	value
Degrees of freedom	6, all revolute
Overall length, fully extended	450.0 mm
Horizontal reach from the J1 axis	360.0 mm
Workspace volume	118 litres
Workspace dead zone	none (measured min radius 0.3 mm)
Task workspace, pen normal to a vertical board	52.3 %
Rated payload	300 g at the TCP
Absolute accuracy (predicted)	1.4 mm RMS, 2.1 mm at 95 %
Repeatability, unidirectional approach	0.70 mm RMS
Structural material	PLA+CF, 2.4 mm flange / 1.6 mm web
Assembled mass	1297 g — see §5, over the 1200 g budget

Actuator — Waveshare ST3215 (datasheet values)

parameter	value
Case	45.22 × 24.72 × 37.25 mm
Output axis position	10.11 mm from one end — NOT centred
Horn disc	Ø19.2 mm
Horn bolt circle	Ø14.0 mm, 4 holes
Stall torque	30 kgf·cm = 2.942 N·m @ 12 V
Locked-rotor current	2.7 A → 32.4 W, all heat
Torque constant k_t	11 kgf·cm/A = 1.079 N·m/A
Encoder	360° magnetic, 4096 counts = 0.0879°
Gear	metal; no backlash figure published

The off-centre output axis is the single most consequential number here. Every servo pocket was originally cut centred on its joint axis, which would have placed every servo

12.50 mm out of position with the horn nowhere near the bearing bore.

Applying the shift then exposed a second problem: the displaced pocket overhung the round housings — by 7.3 mm on the J4 housing and 9.3 mm on the J6 output — so the servo would have stuck out through the wall. Both housings are now racetrack-shaped: a circle at the bearing axis extended along +X to swallow the servo. That costs nothing in the 450 mm chain because it grows perpendicular to it.

housing	was	now	servo containment
wrist_j4_housing	Ø56 circle	65.7 × 54.0 racetrack	2.39 mm margin
wrist_j6_output	Ø52 circle	63.7 × 49.9 racetrack	2.40 mm margin
turret_j1	Ø80 circle	unchanged	4.69 mm margin

Common joint interface

Every joint uses the same three components, so a fit validated on one is valid on all six:

item	specification
Bearing pocket	Ø42.00 mm modelled — gives 0.02 mm interference on a 0.02 mm-shrink printer
Bearing	6806-2RS deep groove, 30 × 42 × 7 — 12 off, C2 preferred
Seat shoulder	Ø38.0 mm, outer race lands on it
Pocket chamfer	0.5 × 45° at the mouth
Shaft	Ø30 mm, steel tube, 5 mm wall
Bearing spacing	22 mm centre-to-centre
Preload	M3 screw drawing the inner races together

2. Joint definition

joint	type	axis	range	servo	worst-case torque	% of stall
J1	revolute	Z (vertical)	±165°	ST3215	0.000 N·m	gravity cannot load it
J2	revolute	Y (pitch)	±115°	ST3215	2.625 N·m	89 %
J3	revolute	Y (pitch)	±150°	ST3215	1.271 N·m	43 %
J4	revolute	Z (roll)	±165°	ST3215	0.296 N·m	10 %
J5	revolute	Y (pitch)	±110°	ST3215	0.297 N·m	10 %
J6	revolute	Z (roll)	±175°	ST3215	0.088 N·m	3 %

All six joints take the same Waveshare ST3215 (30 kgf·cm). The 89 % figure at J2 applies

only to 300 g at full extension; with a pen it falls to 14–29 %.

Link dimensions

segment	length
Base height	50 mm
Base top → J2 (shoulder rise)	40 mm
J2 → J3 (upper arm)	119 mm
J3 → J4 (forearm)	119 mm
J4 → J5	62 mm
J5 → J6	30 mm
J6 → TCP	30 mm
TOTAL	450 mm

3. Fastener schedule

location	fastener	qty	note
Link seam	M3 × 12 into heat-set inserts	12 per link	pitch ≤ 25 mm
Link tip ears	M2.5 into heat-set inserts	2 per link	closes the seam at the bearing
Servo collar clamp	M3 × 16	2 per collar	
Servo case mounting	M2.5	4 per servo	pitch not dimensioned on the outline sheet
Servo horn	M2	4 per joint	Ø14.0 BCD, datasheet
Wrist inter-part	M3	4 per interface	Ø34 bolt circle
Tool flange	M3	4	Ø30 bolt circle
Base foot	M4	4	Ø104 bolt circle
Bearing preload	M3 + spacer	1 per joint	the single most important fastener

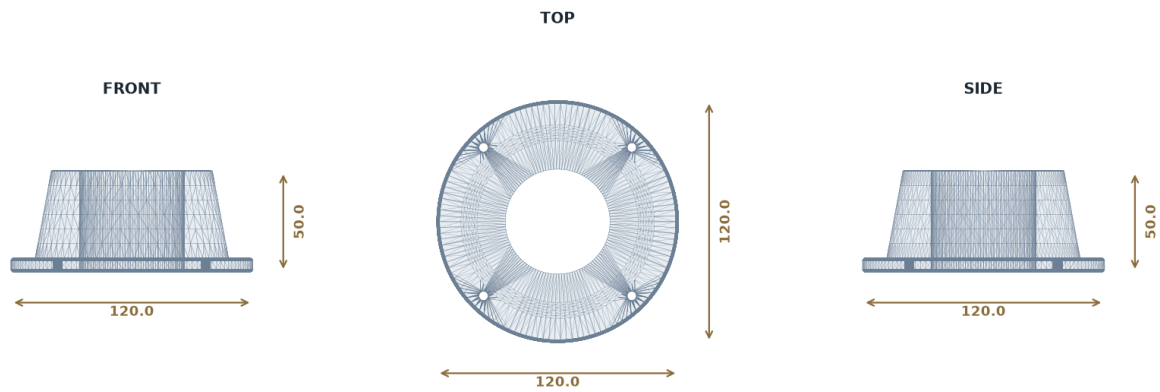
Insert bosses: Ø9.5 outer minimum, Ø4.1 hole, 7.5 mm deep for a Ø4.6 × 5.8 insert.

4. Part drawings

Each sheet carries three orthographic views with overall dimensions and a title block.

ARM-450 · base

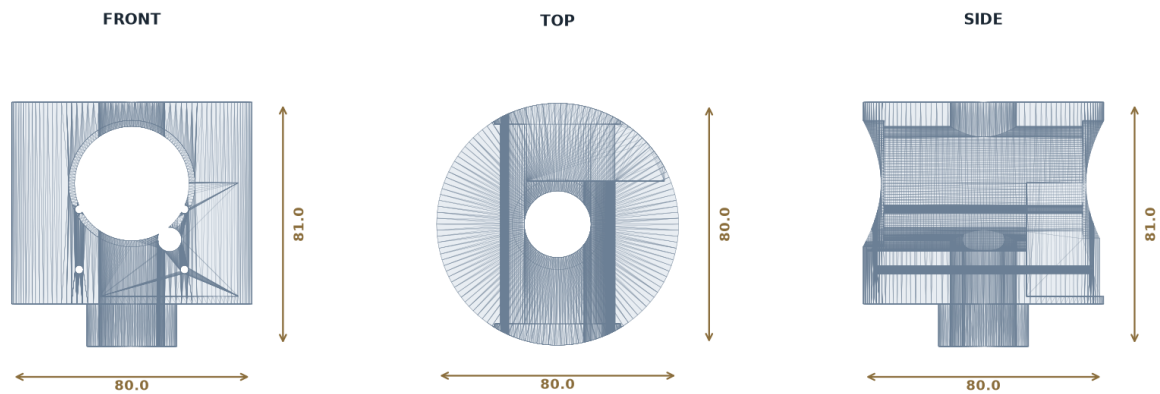
all dimensions mm · 3rd angle



PART	QTY	MATERIAL	ENVELOPE
base	1	PLA+CF	120.0 x 120.0 x 50.0 mm
VOLUME	MASS	NOTE	
234.4 cm3	166.3 g each	HOLLOW to a 3 mm shell — currently modelled solid, 166 g	

ARM-450 · turret_j1

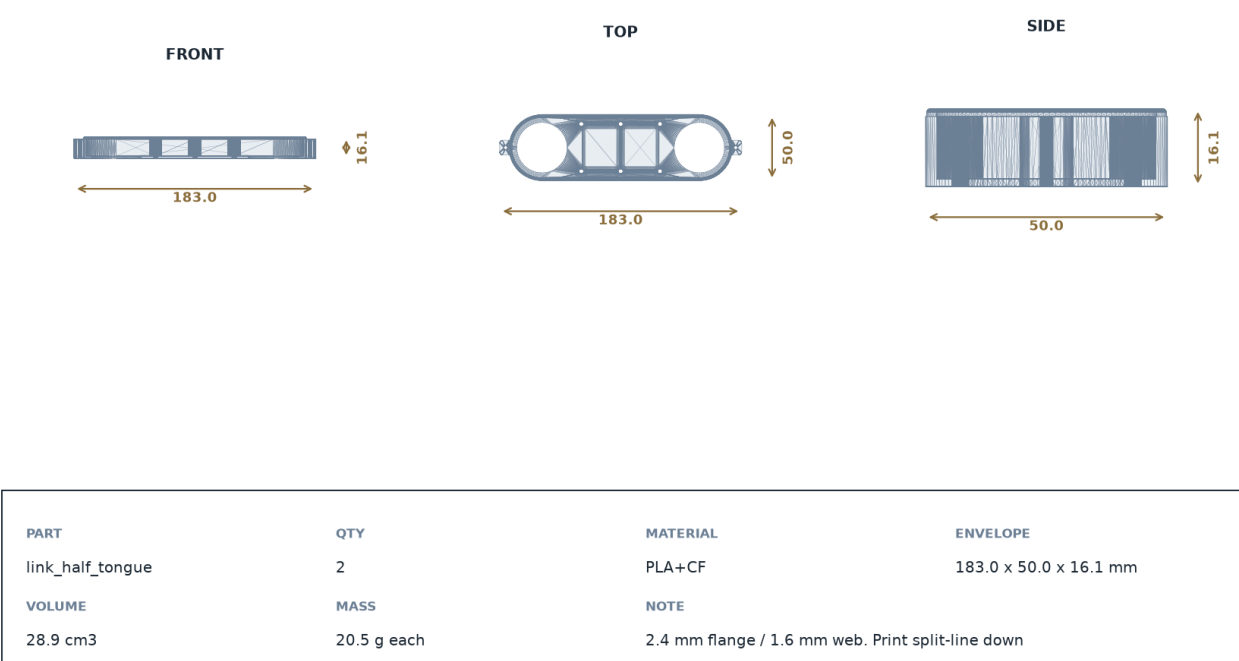
all dimensions mm · 3rd angle



PART	QTY	MATERIAL	ENVELOPE
turret_j1	1	PLA+CF	80.0 x 80.0 x 81.0 mm
VOLUME	MASS	NOTE	
212.4 cm3	150.7 g each	HOLLOW to a 3 mm shell — currently modelled solid, 151 g	

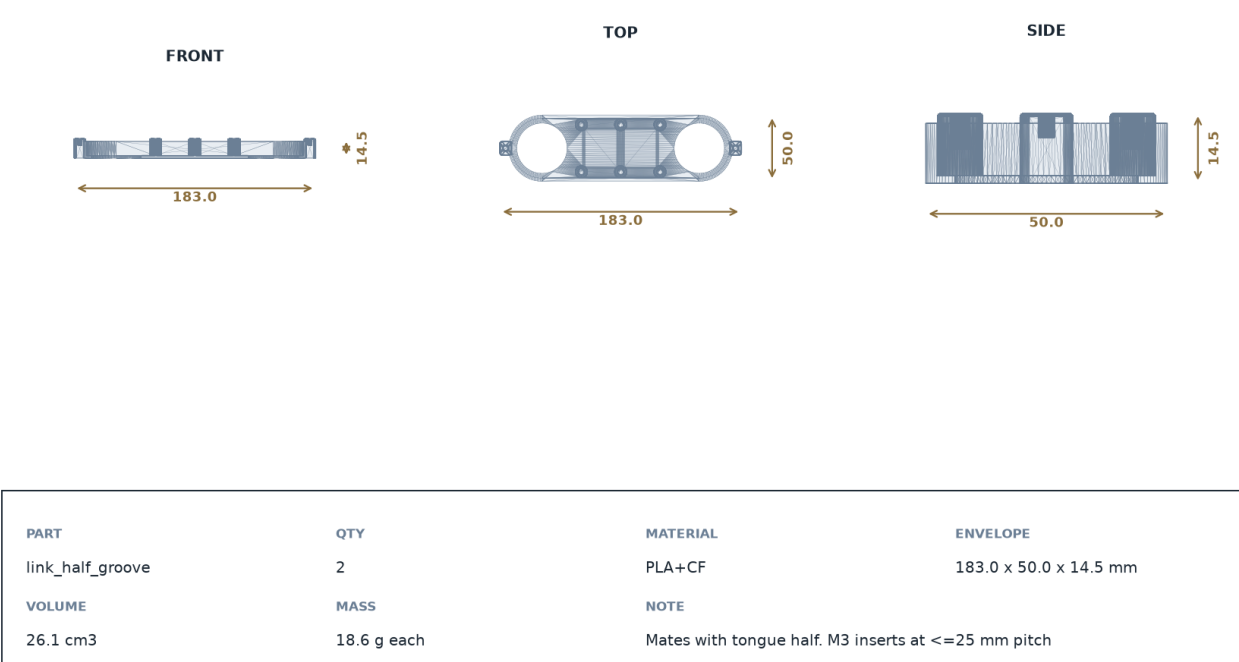
ARM-450 · link_half_tongue

all dimensions mm · 3rd angle



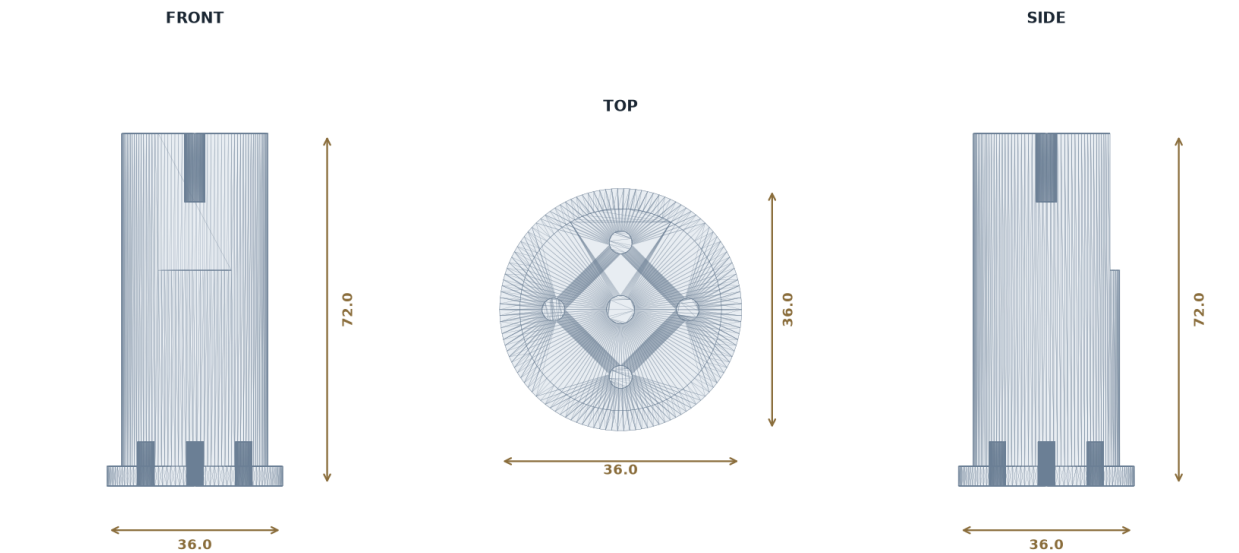
ARM-450 · link_half_groove

all dimensions mm · 3rd angle



ARM-450 · joint_shaft

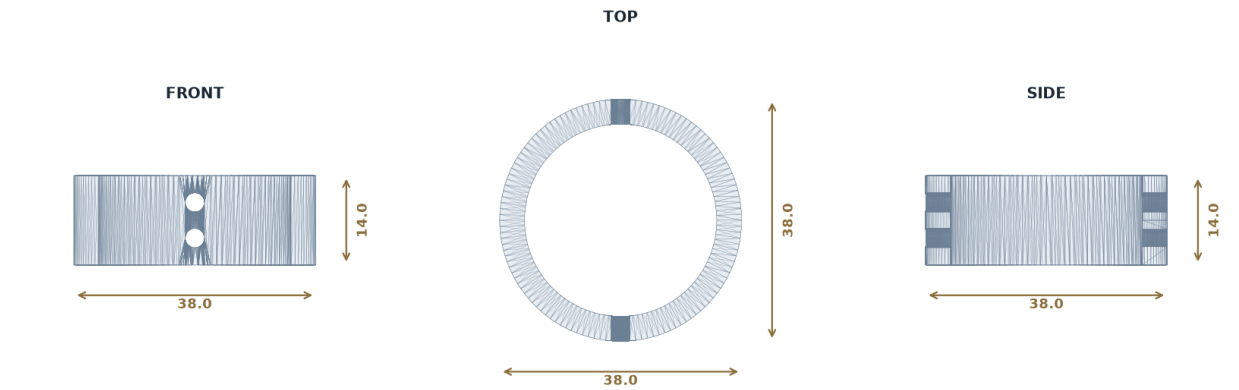
all dimensions mm · 3rd angle



PART	QTY	MATERIAL	ENVELOPE
joint_shaft	2	STEEL — do not print	36.0 x 36.0 x 72.0 mm
VOLUME	MASS	NOTE	
51.1 cm3	36.2 g each	Ø30 x 5 wall tube. Printed PLA creeps at 2.6 N.m	

ARM-450 · shaft_clamp

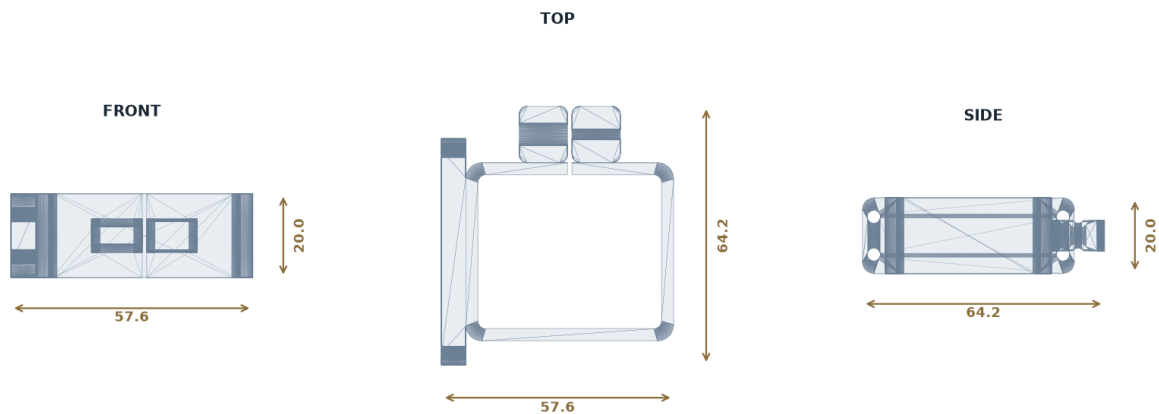
all dimensions mm · 3rd angle



PART	QTY	MATERIAL	ENVELOPE
shaft_clamp	2	PLA+CF	38.0 x 38.0 x 14.0 mm
VOLUME	MASS	NOTE	
5.7 cm3	4.1 g each	Seats in the Ø38 through-bore between the bearings	

ARM-450 · servo_collar

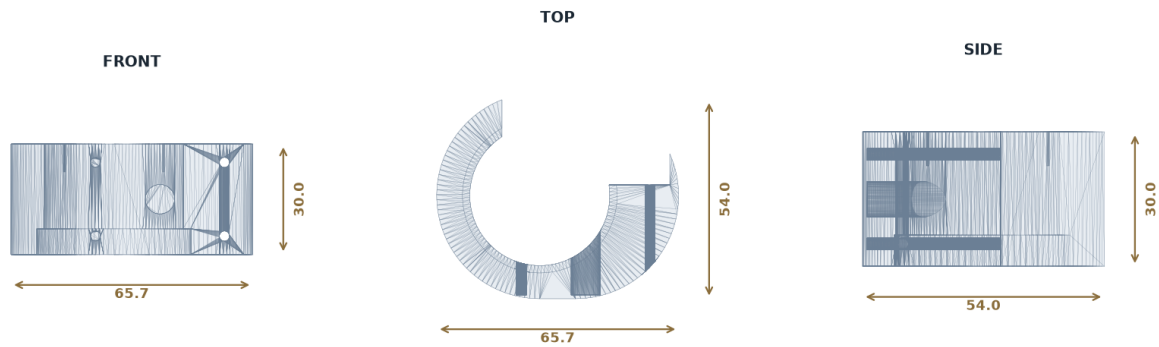
all dimensions mm · 3rd angle



PART	QTY	MATERIAL	ENVELOPE
servo_collar	2	PLA+CF	57.6 x 64.2 x 20.0 mm
VOLUME	MASS	NOTE	
19.4 cm3	13.7 g each	Split clamp, 2 x M3. Bore = servo +0.2 mm/side	

ARM-450 · wrist_j4_housing

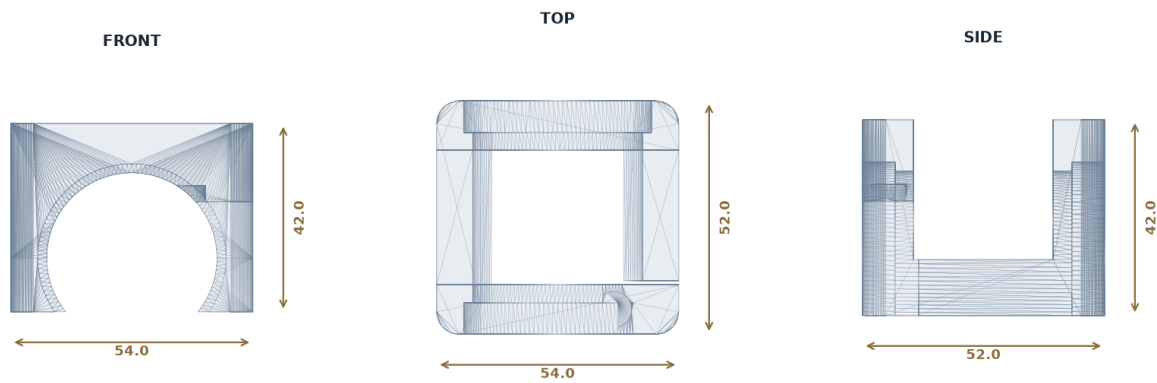
all dimensions mm · 3rd angle



PART	QTY	MATERIAL	ENVELOPE
wrist_j4_housing	1	PLA+CF	65.7 x 54.0 x 30.0 mm
VOLUME	MASS	NOTE	
22.5 cm3	16.0 g each	J4 roll. Ø56 to clear the ST3215	

ARM-450 · wrist_j5_yoke

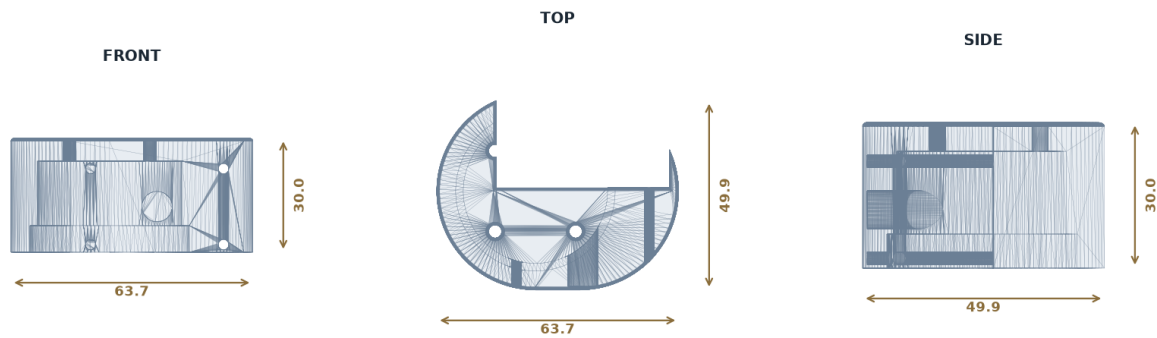
all dimensions mm · 3rd angle



PART	QTY	MATERIAL	ENVELOPE
wrist_j5_yoke	1	PLA+CF	54.0 x 52.0 x 42.0 mm
VOLUME	MASS	NOTE	
29.2 cm3	20.7 g each	J5 pitch fork	

ARM-450 · wrist_j6_output

all dimensions mm · 3rd angle



PART	QTY	MATERIAL	ENVELOPE
wrist_j6_output	1	PLA+CF	63.7 x 49.9 x 30.0 mm
VOLUME	MASS	NOTE	
18.7 cm3	13.3 g each	J6 roll + integral tool flange, 4 x M3 on Ø30 BCD	

5. Mass budget — currently over

This recheck found a problem that earlier estimates missed.

item	qty	mass
6806 bearings	12	384 g
ST3215 servos	6	360 g
base	1	166 g
turret_j1	1	151 g
link halves	4	78 g
joint shafts	2	72 g
wrist parts	3	50 g
collars + clamps	4	36 g
TOTAL		1297 g — 97 g OVER the 1200 g budget

Three fixes

1. Hollow the base and the turret — saves 206 g. Both are currently modelled as solid cylinders. At a 3 mm shell they drop to roughly 35 % of their volume: base 166 → 58 g, turret 151 → 53 g. This is the obvious win and costs nothing.
2. Smaller bearings at the wrist — saves 144 g. J4, J5 and J6 carry only 0.30 N·m; a 6806 there is enormous overkill. A 6704 (20 × 27 × 4) is ~8 g against 32 g. Cost: it breaks the one-bearing-part-number rule — two sizes instead of one.
3. The shaft must be steel, and that costs mass. A printed Ø30 shaft creeps under 2.6 N·m. Solid steel would be ~400 g each, which is far worse; a Ø30 × 5 mm wall tube is ~93 g each, a net +114 g against the printed figure. This is the honest price of a real shaft.

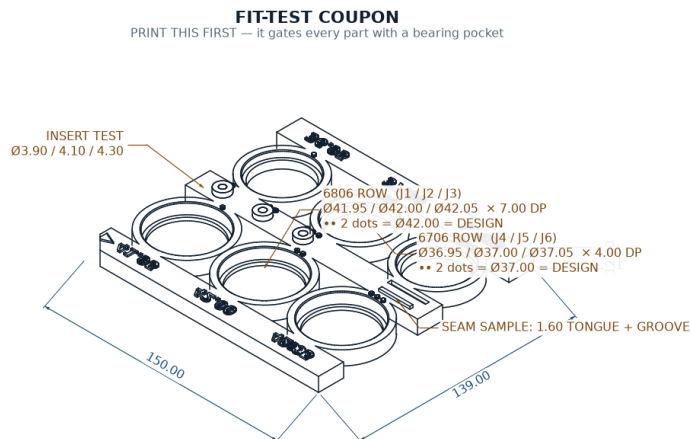
Net: 1297 – 206 – 144 + 114 = 1061 g, inside budget with 139 g of margin.

6. Manufacturing notes

Process	FDM, 0.4 mm nozzle (0.6 mm hardened for PLA+CF)
Layer height	0.16 mm for mounts and brackets, 0.20 mm for shells
Perimeters	6 on mounts, 4 on shells — walls matter far more than infill
Infill	100 % on small brackets, 10-15 % gyroid on shells
Nozzle temperature	225-230 °C PLA, 230-245 °C PLA+CF

Process	FDM, 0.4 mm nozzle (0.6 mm hardened for PLA+CF)
Orientation, link halves	split line on the bed, open face up — support-free, flat mating face
Orientation, bearing bores	axis vertical — a bore built up in layers comes out oval
Printer calibration	measured 0.02 mm XY shrink, Z accurate

6A. Print the fit coupon first



PART	fit_coupon.stl
QTY	1 off
MATERIAL	same filament as the parts it gates
MASS	69.6 g · ~2.3 h
ORIENT	flat, pockets facing UP
ROWS	top Ø42 = 6806, J1/J2/J3 bottom Ø37 = 6706, J4/J5/J6
PURPOSE	2.3 h spent here protects the 19 h of parts that carry a bearing pocket
READ IT	firm thumb press and sits square = right drops in free = too big will not start = too small

fit_coupon.stl — 150 × 65 × 12 mm, 27 g, under an hour. Three bearing pockets stepped 0.05 mm apart (Ø41.95 / Ø42.00 / Ø42.05), three insert-boss hole sizes (Ø3.9 / 4.1 / 4.3) and a tongue-and-groove seam sample.

One print settles all three fits on your machine before eight hours of parts are committed to them. Full instructions in ARM450_BUY.pdf §6.

6B. Bought parts

Everything that is not printed is listed in ARM450_BUY.pdf. The headline: the joints need 12 × 6806-2RS deep-groove bearings (30 × 42 × 7). The Ø42 × 3.7 thrust washer already on the Desktop shares the same outer diameter and will drop into the pocket looking correct, but it has no inner race and zero moment capacity — it is what caused the 28.9 mm wobble.

7. Open items

- Horn bolt circle is now the datasheet Ø14.0 mm, and every servo pocket is shifted

12.50 mm so the output axis lands on the joint axis. The four case screw positions are still estimated — the outline sheet dimensions the case and the horn but not those screws.

- The base and turret are modelled solid and must be hollowed before printing.
- The shaft is drawn as a printed part but must be procured as steel tube.
- 16 % of sampled poses collide with the base at extreme joint angles. All are arm-vs-base, none arm-vs-arm; they bound the usable joint limits rather than indicating a design fault.